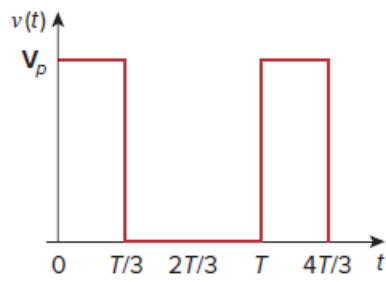


Problem

Using Fig. 11.54, design a problem to help other students better understand how to find the rms value of a waveshape.

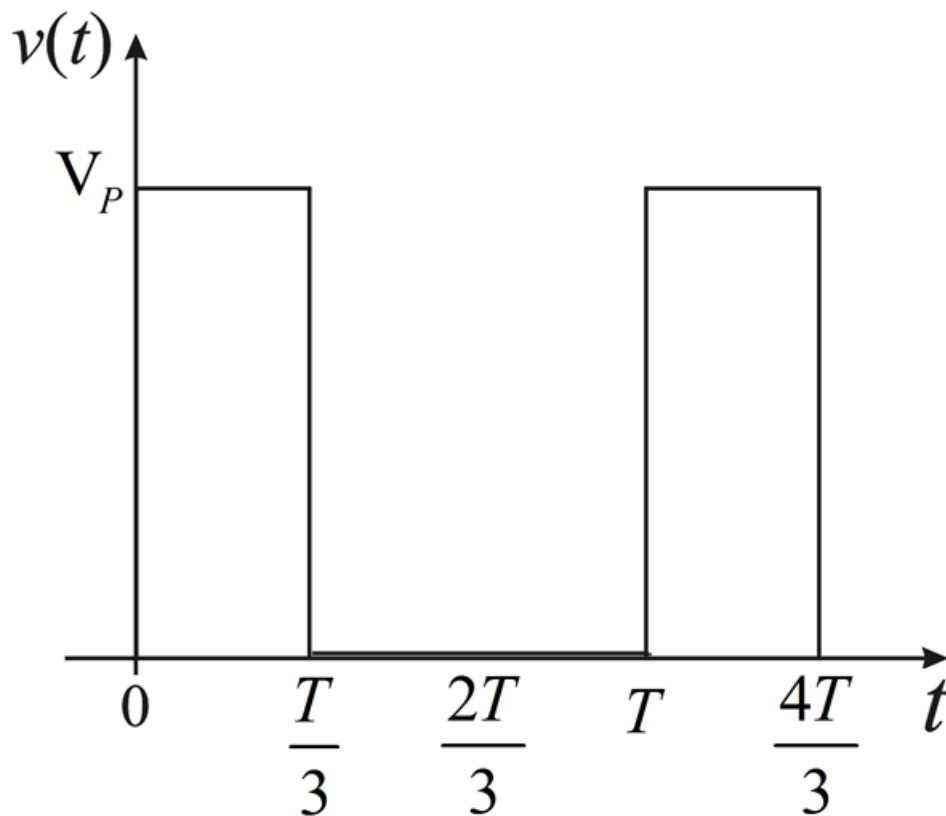
Figure 11.54:



Step-by-step solution

Step 1 of 4

The given voltage waveform is



Step 2 of 4

The period of waveform is T . Over a period, we can write the voltage waveform as

$$v(t) = \begin{cases} V_p, & 0 < t < \frac{T}{3} \\ 0, & \frac{T}{3} < t < T \end{cases}$$

The rms value is

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T [v(t)]^2 dt}$$

Step 3 of 4

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \left[\int_0^{\frac{T}{3}} V_p^2 dt + \int_{\frac{T}{3}}^T 0^2 dt \right]}$$

$$V_{\text{rms}} = \sqrt{\frac{1}{T} V_p^2 \int_0^{\frac{T}{3}} 1 dt}$$

$$V_{\text{rms}} = \sqrt{\frac{1}{T} V_p^2 \left[t \right]_0^{\frac{T}{3}}}$$

Step 4 of 4

$$V_{\text{rms}} = \sqrt{\frac{1}{T} V_p^2 \left[\frac{T}{3} - 0 \right]}$$

$$V_{\text{rms}} = \sqrt{\frac{V_p^2}{3}}$$

$$\therefore \boxed{V_{\text{rms}} = 0.5774 \sqrt{V_p} \text{ V}}$$

Where V_p is the amplitude.